

HEER SHAH

AUTONOMOUS & COMPUTER ENGINEERING | ROBOTICS | AI & DATA ENGINEERING

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UAE Golden Visa Holder

PROFILE

Third-year Autonomous & Computer Engineering (B.Eng. Honours) student at the University of Wollongong in Dubai, building toward autonomous robotic systems powered by emerging technology. Hands-on experience across robotics, embedded systems, AI/ML, generative AI, LLM evaluation, data engineering and engineering research, with enterprise exposure to Azure, Power BI, SharePoint and SAP S/4HANA Cloud. Integrates software, sensors and data-driven methods into working systems, from a GPU-free Raspberry Pi speech-to-sign pipeline to a space debris-tracking platform and a transport platform that cut school-zone congestion by 65%. 1st Place, National AI Hackathon (100+ participants); IEEE conference paper accepted.

EDUCATION

B.Eng. (Honours), Autonomous & Computer Engineering | UOWD, Dubai, UAE Sep 2024 – Sep 2028
• Predicted First Class Honours

Zaka Machine Learning Certification | Capstone: LSTM-based demand forecasting (CashVan) 2026

EXPERIENCE

Research Intern — Data-Driven Human Cardiac Motion Research | Faculty-supervised 2026 – Present
• Analyse human cardiac motion for low-frequency energy harvesting in implantable pacemaker applications.
• Process multi-subject datasets in Python, MATLAB and Google Colab for quantitative and statistical analysis, extracting displacement, velocity, acceleration and frequency characteristics.

Robotics & Automation Intern | CodeAlpha Aug 2026
• Developed robotics and automation solutions involving software implementation, hardware/software integration, control logic, testing and troubleshooting.
• Applied programming and engineering problem-solving to integrate physical and software components.

SAP Dual Studies Program Participant | SAP Jul 2026
• Completed training in SAP S/4HANA Cloud Public Edition, enterprise processes and digital business systems.

AI Intern & Zaka AI Ambassador | Zaka AI, Dubai Jul 2025 – Sep 2025
• Evaluated and optimized LLM applications using Promptfoo and Opik, analysing model performance, reliability and evaluation results.
• Built an enterprise AI solution and Azure chatbot and analytics prototypes integrating Power BI and SharePoint for data-driven business workflows.

SELECTED PROJECTS

Intelligent Transport Platform | Bounty Challenge, UOWD with RTA Jan 2026
• Developed a software-only platform addressing school transportation inefficiencies without infrastructure changes.
• Implemented real-time dashboards, live bus tracking and AI-driven scheduling for parents, schools and RTA stakeholders.
• Achieved reported reductions of 65% in school traffic congestion and 78% in bus delays.

LENZO Avatar — Real-Time Speech-to-Sign Language System | Engineering Project May 2026
• Built a GPU-free pipeline running locally on a Raspberry Pi, integrating Vosk speech recognition, NLP, intent classification, sign-gloss generation, SiGML, WebSockets and browser-based 3D visualization.
• **Bronze Medal** award.

HERA — AI Space Domain Awareness System | Project Lead • Supernova finalist (top 10) 2025
• Led development of a platform for satellite tracking, debris detection and collision prediction using live NASA and CelesTrak data.
• Integrated APIs, data processing, AI analysis and real-time visualization.

- Developed an AI-powered household energy forecasting and recommendation system with Arabic-language support.
- **1st Place** — National AI Hackathon, 100+ participants.

CashVan — Inventory Intelligence System

2026

- Processed 1,067,371 retail transactions: handled 247,389 missing values and 34,335 duplicate records, producing 779,425 cleaned records.
- Built LSTM and Prophet demand-forecasting models with preprocessing and feature engineering (MAE of 2,561.82 and 4,376.15).
- **ASET IEEE conference paper accepted.**

TECHNICAL SKILLS

Robotics & Autonomy: Autonomous Systems, Robotics, Embedded Systems, Raspberry Pi, Sensors, Control, Hardware/Software Integration

AI & Machine Learning: Python, TensorFlow/Keras, Scikit-Learn, NLP, Computer Vision, Generative AI, Predictive Modelling, LLM Evaluation, Promptfoo, Opik

Data Engineering & Analytics: Pandas, NumPy, MATLAB, Google Colab, Data Pipelines, Data Cleaning, Statistical Analysis, Forecasting

Cloud & Enterprise: Microsoft Azure, Power BI, SharePoint, SAP S/4HANA Cloud

Development: C++, JavaScript, TypeScript, React, Flask, REST APIs, Git/GitHub, DSA

Areas of Interest: Emerging technology for autonomous systems, robotics, space systems, intelligent infrastructure

ACHIEVEMENTS

- **1st Place** — Microsoft & Zaka AI National Hackathon (ETRNA), 100+ participants
- **IEEE Conference Paper, Accepted** — Advanced Science and Engineering Technology (ASET), CashVan
- **Bronze Medal** — LENSO
- **Silver Medal** — Robotics Workshop, recognized for building and programming a functional autonomous robot